

SMART HOSTEL MANAGEMENT SYSTEM

Java-Based Object-Oriented Programming Project Report

1. Abstract

The Smart Hostel Management System is a Java-based console application developed to simplify and organize basic hostel management activities. The system allows users to register students, add rooms, allocate and vacate rooms, submit and resolve complaints, and calculate hostel fees. The project demonstrates Object-Oriented Programming concepts such as encapsulation, inheritance, polymorphism, abstraction, interfaces, constructors, ArrayList, exception handling, and file handling. Student, room, and complaint information can be stored in text files and loaded when the application is restarted.

2. Introduction

Hostel management involves maintaining student records, room information, occupancy, complaints, and fee details. Managing these activities manually can become difficult as the number of students and rooms increases. This project provides a simple computerized solution through a menu-driven Java application and demonstrates how Java OOP concepts can be applied to a practical real-world problem.

3. Problem Statement

Manual hostel administration can make it difficult to maintain accurate student records, track room availability, manage complaints, and preserve information. The proposed system provides a structured application for performing these operations using Java.

4. Objectives

- Register and manage student information.
- Add and manage single and shared hostel rooms.
- Allocate and vacate rooms while checking availability.
- Submit and resolve maintenance and electrical complaints.
- Calculate hostel fees according to room type.
- Save and load hostel information using text files.
- Demonstrate important Java OOP concepts.

5. Technologies Used

Java, Visual Studio Code, ArrayList, text-file storage, file handling, and exception handling.

6. OOP Concepts Used

Encapsulation is implemented using private data members with getters and setters. Inheritance is demonstrated through Student extending Person and room subclasses extending Room. Polymorphism is

demonstrated through method overriding. Abstraction is represented by the FeeCalculable interface. Constructors are used to initialize objects.

7. Main Modules

Student Management, Room Management, Complaint Management, Fee Management, and File Management.

8. Class Structure

The project is divided into separate classes with clear responsibilities. Person contains common personal information, Student extends Person, and SingleRoom and SharedRoom extend Room. Complaint is the base class for MaintenanceComplaint and ElectricalComplaint. Hostel coordinates the main operations, while FileManager and InputHelper handle file operations and input validation.

9. Working of the System

When the application starts, previously saved student, room, and complaint records are loaded. The main menu is displayed and the user selects an operation. After completing an operation, the system returns to the main menu until Exit is selected. The Save Data option stores current records in the data folder.

10. Output Screenshots

The following screenshots show the working of the implemented modules during testing.

```

=====
SMART HOSTEL MANAGEMENT
=====
1. Register Student
2. Add Room
3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit
=====
Enter your choice: 1

===== REGISTER STUDENT =====
Enter Student ID: 101
Enter Student Name: Pratibha Jha
Enter Phone Number: 7488987646
Enter Course: CSE AI ML
Student ID already exists!

=====
SMART HOSTEL MANAGEMENT
=====
1. Register Student
2. Add Room
3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit
=====
Enter your choice: 2

===== ADD ROOM =====
Enter Room Number: 228

Select Room Type:
1. Single Room
2. Shared Room
Enter your choice: 2
Enter Room Capacity: 3
Room added successfully!

```

Figure 1: Student registration, duplicate ID validation, and room addition

```

=====
      SMART HOSTEL MANAGEMENT
=====
1. Register Student
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3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit
=====
Enter your choice: 3

===== ALLOCATE ROOM =====
Enter Student ID: 101
Enter Room Number: 228
Student already has a room!

=====
      SMART HOSTEL MANAGEMENT
=====
1. Register Student
2. Add Room
3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit
=====
Enter your choice: 4

===== VACATE ROOM =====
Enter Student ID: 101
Room vacated successfully!

```

Figure 2: Room allocation validation and successful room vacation

Enter your choice: 5

===== STUDENT LIST =====

Student ID: 101

Name: Pratibha

Phone: 76324876856

Course: CSE

Room: Not Allocated

=====

SMART HOSTEL MANAGEMENT

=====

1. Register Student
2. Add Room
3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit

=====

Enter your choice: 6

===== ROOM LIST =====

Room Type: Shared Room

Room Number: 209

Capacity: 3

Occupied: 0

Figure 3: Student and room details displayed by the system

Enter your choice: 7

===== SUBMIT COMPLAINT =====

Enter Complaint ID: 34

Enter Student ID: 101

Select Complaint Type:

1. Maintenance

2. Electrical

Enter your choice: 2

Enter Complaint Description: Fan Not Working

Complaint submitted successfully!

=====

SMART HOSTEL MANAGEMENT

=====

1. Register Student

2. Add Room

3. Allocate Room

4. Vacate Room

5. View Students

6. View Rooms

7. Submit Complaint

8. View Complaints

9. Resolve Complaint

10. Calculate Hostel Fee

11. Save Data

0. Exit

=====

Enter your choice: 8

===== COMPLAINT LIST =====

Complaint Type: Electrical

Complaint ID: 302

Student ID: 101

Description: Fan Not working.

Status: Pending

Figure 4: Complaint submission and complaint list

```
Enter your choice: 9

===== RESOLVE COMPLAINT =====
Enter Complaint ID: 34
Complaint resolved successfully!

=====
                SMART HOSTEL MANAGEMENT
=====

1. Register Student
2. Add Room
3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit

=====

Enter your choice: 10

===== HOSTEL FEE =====
Enter Student ID: 101
Student has no room allocated.
```

Figure 5: Complaint resolution and fee validation for an unallocated student

```
=====
Enter your choice: 11

===== SAVING DATA =====
Student data saved successfully!
Room data saved successfully!
Complaint data saved successfully!
All data saved successfully!

=====
                SMART HOSTEL MANAGEMENT
=====

1. Register Student
2. Add Room
3. Allocate Room
4. Vacate Room
5. View Students
6. View Rooms
7. Submit Complaint
8. View Complaints
9. Resolve Complaint
10. Calculate Hostel Fee
11. Save Data
0. Exit

=====
Enter your choice: 0

=====
Thank you for using Smart Hostel!
=====
PS C:\Users\jhapr\OneDrive\Desktop\SmartHostelManagement>
```

Figure 6: Data saving and successful program exit

11. Testing

Test Case	Expected Result	Status
Register new student	Student added successfully	Pass
Duplicate student ID	Duplicate rejected	Pass
Add room	Room added successfully	Pass
Already allocated student	Allocation rejected	Pass
Vacate room	Room vacated successfully	Pass
Submit complaint	Complaint added	Pass
Resolve complaint	Complaint resolved	Pass
Fee without room	No room allocated message	Pass

Save data	Data saved successfully	Pass
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12. Advantages

- Simple console interface.
- Organized student and room records.
- Room availability checking.
- Complaint tracking and resolution.
- Persistent storage using text files.

13. Limitations

The current version is console-based, uses text files instead of a database, and does not include authentication, online payments, or notifications.

14. Future Scope

The project can be expanded with a graphical interface, MySQL database, login system, online fee payment, receipt generation, search and filtering, notifications, and hostel statistics.

15. Conclusion

The Smart Hostel Management System demonstrates the practical use of Java and OOP concepts. It provides basic hostel operations including student registration, room management, allocation, complaint handling, and fee calculation. The project also provides experience with collections, exception handling, and file handling.

16. References

- Oracle Java Documentation.
- Java Platform, Standard Edition Documentation.
- Course material for Programming in Java.
- Visual Studio Code Documentation.